

### Subject Description Form

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| <b>Subject Code</b>                         | AF3213                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Subject Title</b>                        | Business Analytics in Accounting and Finance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Credit Value</b>                         | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Level</b>                                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Pre-requisite/Co-requisite/Exclusion</b> | Pre-requisites:<br>Introduction to Statistics for Business (AMA1501) or equivalent and<br>Financial Accounting (AF2108) or equivalent and<br>Management Accounting 1 (AF2110) and<br>Introduction to Business Analytics (LGT2425/MM2425) or Business Analytics (LGT3425/MM3425)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Objectives</b>                           | This subject introduces students to the basic concepts, methods, and approaches of data analytics in accounting and finance. The growing volume of both structured and unstructured data has pushed forward a more data-driven form of decision-making. Future accountants and financial analysts need to be able to collect and work with data. This subject will equip students with work-related data analytic skills and mindsets for real-world business scenarios. The contents include (1) knowledge about various useful accounting and financial datasets, (2) skills in acquiring, processing, analyzing, and visualizing data, (3) experiences in producing real-world business analytical reports and generating business insights, and (4) norms of code and data management and workplace communications. |
| <b>Subject Learning Outcomes</b>            | Upon completion of the subject, students will be able to:<br>a) Know how data analytics drive decisions in accounting and finance contexts.<br>b) Understand how to summarize accounting and financial data.<br>c) Analyze and present data sets using accounting and finance cases.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Subject Synopsis/Indicative Syllabus</b> | <b>Basic Concepts and Methods of Data Analytics</b><br>Data retrieving; Data preprocessing and cleaning; Data analytics approaches; Data visualization and summarization.<br><b>Applications of Data Analytics in Accounting and Finance</b><br>Diagnostic, predictive, and prescriptive analytics in various projects in accounting and finance.<br><b>Data Visualization</b><br>Communicate results effectively using clear structure, language, and visualizations.                                                                                                                                                                                                                                                                                                                                                  |
| <b>Teaching/Learning Methodology</b>        | Seminars and in-class and after-class exercises will introduce key concepts and techniques. The subject places a lot of emphasis on project work. Students will be required to deliver a project that emphasizes real-world accounting and finance issues. By completing the project, students should have hands-on experience using the knowledge they have learned in class to solve accounting and finance problems. Students are encouraged to actively share their views and experiences with their lecturer and classmates.                                                                                                                                                                                                                                                                                       |

| Assessment Methods in Alignment with Intended Learning Outcomes | Specific assessment methods/tasks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | % weighting | Intended subject learning outcomes to be assessed (Please tick as appropriate) |   |   |  |          |  |
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|                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |             | a                                                                              | b | c |  |          |  |
|                                                                 | <b>Continuous Assessment</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |             |                                                                                |   |   |  |          |  |
|                                                                 | 1. Attendance & Class Participation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 10%         | ✓                                                                              | ✓ | ✓ |  |          |  |
|                                                                 | 2. In-Class Quizzes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 10%         | ✓                                                                              | ✓ | ✓ |  |          |  |
|                                                                 | 3. Lab Assignments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 30%         | ✓                                                                              | ✓ | ✓ |  |          |  |
|                                                                 | 4. Project Report & Presentation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 50%         | ✓                                                                              | ✓ | ✓ |  |          |  |
|                                                                 | Total                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 100%        |                                                                                |   |   |  |          |  |
|                                                                 | <p>Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:</p> <ul style="list-style-type: none"> <li>● Attendance &amp; Class Participation: Students need to attend the scheduled classes. The attendance will be recorded each week through class participation. Class participation is an important activity for students' learning experience. Therefore, students' participation is highly valued. Participation involves contributing meaningful content related to the learning materials for the class, including answering questions during class and discussing issues with classmates through discussion forums.</li> <li>● In-Class Quizzes: Students complete in-class online quizzes, assessing their understanding of the key concepts.</li> <li>● Lab Assignments: Lab assignments test students' overall ability to apply the knowledge they have learned in the subject.</li> <li>● Project Report and Presentation: These are used to test students' ability to understand the materials and to achieve the intended learning outcomes through a more in-depth investigation of different topics.</li> </ul> |             |                                                                                |   |   |  |          |  |
| Student Study Effort Expected                                   | Class contact:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |             |                                                                                |   |   |  |          |  |
|                                                                 | Seminars                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |             |                                                                                |   |   |  | 39 Hrs.  |  |
|                                                                 | Other student study efforts:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |             |                                                                                |   |   |  |          |  |
|                                                                 | Reading materials/textbook questions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |             |                                                                                |   |   |  | 39 Hrs.  |  |
|                                                                 | On average, around 16 hours will be spent on the assignment quizzes and around 20 hours on the group project discussion, presentation, and written report                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |             |                                                                                |   |   |  | 36 Hrs.  |  |
|                                                                 | Total student study effort                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |             |                                                                                |   |   |  | 114 Hrs. |  |

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| <b>Reading List and References</b> | <i>Introduction to Data Analytics for Accounting</i> , 2023, by Richardson, Teeter and Terrell, McGraw-Hill. 2 <sup>nd</sup> Edition, McGraw Hill. |
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